

SAI GOPALA SWAMY GADDE

Software Engineer | Full-Stack | Cloud | GenAI / LLM | Machine Learning

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 [LinkedIn](#) |  Dallas, TX

PROFESSIONAL SUMMARY

Innovative **Software Engineer with 3+ years of experience** designing **cloud-native, AI-driven, and full-stack applications** for enterprise clients including **DIRECTV and Virtusa**.

Specialized in **Generative AI, LLM optimization, prompt engineering, fine-tuning, embedding pipelines, and model deployment**. Delivered impactful AI solutions by **enhancing operational efficiency, improving response quality, and building scalable inference systems**.

Proficient in **Python, React, Node.js, AWS Cloud, TensorFlow, Keras, LangChain, RAG, Pinecone, MLflow, and Weights & Biases**. Passionate about bridging engineering precision with AI-driven innovation.

PROFESSIONAL EXPERIENCE

Software Engineer – AI & Cloud – DIRECTV

El Segundo, CA | **Sep 2023 – Present**

Tech: Python, Node.js, React, AWS (Lambda, EC2, S3, RDS, CloudWatch), Docker, LangChain, OpenAI API, Pinecone, MLflow, W&B

Roles & Responsibilities

- Developed **GenAI-driven customer experience features** using **OpenAI GPT-4, LangChain, and Pinecone embeddings** to improve operational efficiency and accuracy.
- Built and deployed **embedding models** (text-embedding-3-large, sentence transformers) for semantic search and knowledge retrieval.
- Implemented **RAG pipelines** enabling real-time knowledge augmentation and enhanced AI responses.
- Designed internal frameworks for **prompt engineering, model optimization, and evaluation**.
- Deployed **AI microservices on AWS Lambda and EC2**, containerized with Docker for streamlined CI/CD and monitoring.
- Built **LLM optimization scripts** and prompt engineering frameworks to standardize and evaluate model outputs.
- Designed internal dashboards and monitoring tools for AI inference performance, latency, and prompt evaluation metrics.

Software Engineer – Virtusa Consulting Services

Hyderabad, India | **Apr 2022 – Jun 2023**

Tech Stack: Python, Java, Flask, AWS, Docker, REST APIs, SQL, Cloud Automation

Roles & Responsibilities

- **Built RESTful APIs and data pipelines supporting AI-enhanced services.**

- **Assisted in designing cloud architectures and** microservice patterns, **improving system reliability and scalability.**
- **Integrated** ML model tracking & evaluation using MLflow and Weights & Biases (W&B) **for AI prototypes.**
- **Automated deployments and MLOps workflows using** Docker and AWS CLI.
- **Contributed to vector-based semantic search systems and internal AI tooling.**

PROJECTS

Alzheimer's Disease Prediction – Hybrid IV3–VGG19 (AI / Deep Learning / Computer Vision)

- Built a hybrid deep learning model for MRI-based Alzheimer's classification.
- Developed a Tkinter GUI for **real-time model inference.**
- Applied preprocessing, augmentation, and feature extraction for robust predictions.

Predictive Flight Delay Analysis (Machine Learning / AI)

- Developed **ML pipelines** and predictive models for airline schedule optimization.
- Built dashboards and visualizations to provide operational insights.

IoT-Based Manhole Alert System (Patent Pending)

- Built IoT safety solution using sensors, GPS & Firebase with a community reporting app.
- Patent application filed for smart-city hazard mapping system.

EDUCATION

Master of Science in Computer Science

University of Texas at Arlington | 2023–2025 | CGPA: 3.8

Bachelor of Technology in Computer Science Engineering

VR Siddhartha Engineering College | 2017–2021

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, SQL

Frontend: React.js, Redux, HTML/CSS, JavaScript

Backend & APIs: Node.js, Express.js, Flask, REST APIs, JWT

GenAI & ML: OpenAI APIs, LangChain, RAG, Pinecone, Llama Models, TensorFlow, Keras, Scikit-learn

Cloud: AWS (Lambda, EC2, S3, RDS, IAM), Azure, GCP, Firebase

DevOps: Docker, GitHub Actions, CI/CD, CloudWatch

Databases: PostgreSQL, MongoDB, MySQL

Testing: Jest, Pytest, Selenium, React Testing Library

CERTIFICATIONS

AZ-900 (Azure Fundamentals) | AI-900 (Azure AI Fundamentals) | Cisco DevNet Associate | PCAP – Python | CPA – C++ | CLA – C | Intro to Cybersecurity | Intro to IoT | NDG Linux Unhatched | R Programming (NPTEL) | Python for Everybody (Michigan) | AI For Everyone (DeepLearning.AI)